

# Report R11 — One rule space, two gates

What changes, and what cannot change, when the Hadamard is replaced by  $X$

QCA Sector Analysis project (Tier 1)

2026-07-31 (R9 1e.1 vs R10 1x.1, obc0, common window  $N \leq 16$ )

## Abstract

R1–R9 analyse 256 brick-wall rules whose active gate is the Hadamard; R10 analyses the same 256 symbol tables with the gate changed to  $X$ . This report compares them. The two families turn out not to be independent: because applying the Hadamard to a site produces *both* values of the target bit, the flip is always one of the outcomes, so the  $X$  transition graph is a subgraph of the Hadamard one and the  $X$  sectors **refine** the Hadamard sectors — verified exhaustively for all 256 rules and over 2816 swept units, with zero exceptions. Three consequences organise everything else. The gate can only ever *split* sectors, never merge them, so no rule loses sector structure: of the 51 rules with an exponentially growing sector count under  $H$ , all 51 keep one under  $X$ , and 39 rules with no structure under  $H$  acquire it. The 81 rules with no  $V$  are literally the same circuit under both gates, which makes the two independent engines a cross-implementation test that they pass on all 916 shared units. And on the 16-rule unitary/reversible baseline — the only fully like-for-like comparison, since both families satisfy sectors = attractors there — the gate change is nearly a reflection: 9 of the 11 rules that have a single giant sector under  $H$  end up with no giant sector at all under  $X$ , 7 of them at the opposite extreme of the axis, with rule 204 the sole fixed point and rule 51 going from a single sector of  $2^N$  states to  $2^{N-1}$  two-cycles. Quantitatively almost nothing transfers: on the unitary baseline the Pearson correlation between the two gates' exponents is +0.17 in  $a$  and +0.03 in  $b$ . obc0 only; pbc is held for both families.

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# 1 What is being compared, and on what window

The rule encoding is shared exactly. A rule is a tuple  $(r_{00}, r_{01}, r_{10}, r_{11})$  over  $\{\mathbf{I}, \mathbf{V}, \mathbf{D}, \mathbf{E}\}$ ; the circuit applies the symbol selected by a site's two neighbours, brick-wall, even sublattice ascending and then odd, each reading the current state. Both engines compile the same step list from `core/cycle.py`. The single difference is what  $\mathbf{V}$  means:

| symbol       | Hadamard family (R1–R9) | permutation family (R10) |
|--------------|-------------------------|--------------------------|
| $\mathbf{I}$ | identity                | identity                 |
| $\mathbf{V}$ | Hadamard $H$            | $X$                      |
| $\mathbf{D}$ | reset to $ 0\rangle$    | reset to $ 0\rangle$     |
| $\mathbf{E}$ | reset to $ 1\rangle$    | reset to $ 1\rangle$     |

**The window.** R9 fits its bases on  $N \leq 16$  and R10 on  $N \leq 24$ . A base fitted on a longer window is a different number, so every comparison below rebuilds the  $X$ -gate descriptors on **the same window as the Hadamard ones**,  $N \leq 16$ , through the identical `scaling/sectors.py` machinery. Nothing is compared across windows. Where an  $N = 24$  number is quoted it is labelled. One consequence is visible in the tables: rule 108's  $X$ -gate cycle length is classified *linear* on  $N \leq 16$  and *exponential* on  $N \leq 24$ , which is a statement about how much data the classifier has, not about the rule.

**What is not compared.** The Hadamard family carries objects with no counterpart here — dark states, the monitored/unmonitored certificate of R7, the protected-coherence gap — because they need interference. Conversely the permutation family's transient structure has no Hadamard counterpart worth plotting. The comparison is restricted to the two coordinates both families define: the sector count  $n_{\text{wcc}} \sim a^N$  and the largest sector  $D_{\text{max}} \sim b^N$ , plus the growth *class* of each.

## 2 The two families are nested

The comparison is not between two unrelated data sets. There is an exact containment, and it is elementary.

**Theorem 1** (refinement). *For every rule, every  $N$  and both boundary conventions,  $\text{succ}_X(x) \in \text{succ}_H(x)$  for every basis state  $x$ . Hence the  $X$  transition graph is a subgraph of the Hadamard transition graph on the same vertex set, and the  $X$  sectors refine the Hadamard sectors: every  $X$ -component lies inside a single  $H$ -component, and every Hadamard sector is a disjoint union of  $X$  sectors.*

*Proof.* Consider one elementary step. At  $\mathbf{I}$ ,  $\mathbf{D}$ ,  $\mathbf{E}$  sites the two families do the same thing. At a  $\mathbf{V}$  site the Hadamard maps  $|b\rangle$  to  $(|0\rangle \pm |1\rangle)/\sqrt{2}$ , so the measured or Kraus-resolved successor set contains *both* values of the target bit; the  $X$  action, which sets it to  $1 - b$ , is one of them. The circuit is a composition of such steps, and choosing the flip at every  $\mathbf{V}$  site along the sweep is one branch of the Hadamard successor set. So each  $X$  edge is an  $H$  edge. Union-find over a subgraph produces a refinement.  $\square$

**Corollary 1.**  $n_{\text{wcc}}^X(N) \geq n_{\text{wcc}}^H(N)$  and  $D_{\text{max}}^X(N) \leq D_{\text{max}}^H(N)$ , pointwise in  $N$ ; asymptotically  $a_X \geq a_H$  and  $b_X \leq b_H$ .

**Corollary 2.** *A rule with no  $\mathbf{V}$  never invokes the gate, so its two circuits are the same map. There are 81 such rules (204 and the 80  $\mathbf{V}$ -free ones), and for them the refinement is an equality.*

All of this is verified rather than assumed:

| check                                    | window         | holds     | unit              |
|------------------------------------------|----------------|-----------|-------------------|
| $\text{succ}_X(x) \in \text{succ}_H(x)$  | $N = 8, N = 9$ | 256/256   | rules, all states |
| X sectors refine H sectors               | $N = 8, N = 9$ | 256/256   | rules, exhaustive |
| $n_{\text{wcc}}^X \geq n_{\text{wcc}}^H$ | $N \leq 16$    | 2816/2816 | units             |
| $D_{\text{max}}^X \leq D_{\text{max}}^H$ | $N \leq 16$    | 2816/2816 | units             |
| V-free rules identical in both stores    | all $N$        | 916/916   | units, 81 rules   |

The last row is worth its own sentence. The Hadamard sectors come from a set-valued successor relation streamed through union-find; the X-gate sectors come from a numpy step table and a three-colour walk. The two share no code below `core/cycle.py::_compile`. On the 81 rules where they must agree they agree exactly — the full sector-size multiset, at every  $N$  where both were run. That is the strongest cross-implementation check either report has.

**Reading the theorem physically.** The Hadamard’s extra edge is the non-flip branch, and it is exactly what *glues*  $X$  sectors together. So one may think of the Hadamard sector structure as the  $X$  sector structure after gluing: superposition is a coarsening operation on the sector lattice. That is the reverse of the intuition one might start with — that a superposing gate would *create* structure — and it explains the census in the next section without any further input.

### 3 No rule loses its sector structure

Corollary 1 says the sector count cannot fall. The growth-class cross-tabulation makes the consequence concrete:

| under $H$   | under $X$ |        |             |           |     |
|-------------|-----------|--------|-------------|-----------|-----|
|             | constant  | linear | exponential | irregular | all |
| constant    | 134       | 6      | 28          | 5         | 173 |
| polynomial  | 0         | 20     | 10          | 0         | 30  |
| exponential | 0         | 0      | 51          | 0         | 51  |
| irregular   | 0         | 0      | 0           | 2         | 2   |
| all         | 134       | 26     | 89          | 7         | 256 |

Every entry below the diagonal is zero, as it must be. Reading the rows:

- **All 51 rules with an exponential sector count under the Hadamard keep one under  $X$ .** Hilbert-space fragmentation, measured by the growth class of the sector count, is not a property of the Hadamard: it survives replacing the superposing gate by a permutation.
- 39 of the 173 rules with a *constant* sector count under  $H$  — no structure at all, one or two sectors covering the space — acquire structure under  $X$ : 28 exponential, 6 linear, 5 irregular. The gate change can only add.
- 10 of the 30 rules with a linear sector count under  $H$  become exponential; the other 20 stay linear.

The reverse reading is the one that matters for the Hadamard reports: if a rule has no sector structure under  $X$ , it has none under  $H$  either. The 134 rules in the top-left cell are structureless under both gates, and that is a property of their symbol table rather than of the dynamics.

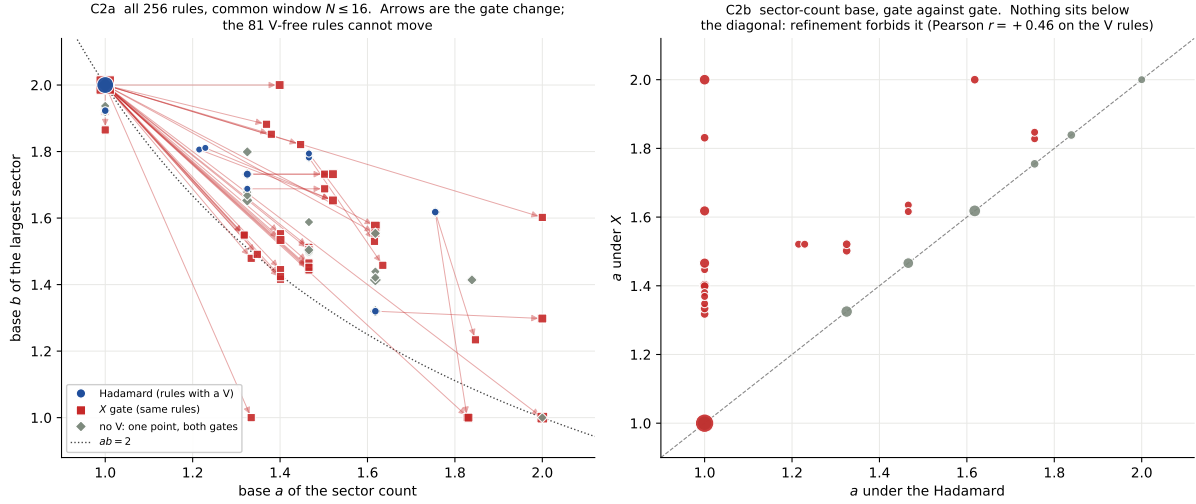


Figure 1: **C2, all 256 rules on a common window.** Left: the sector plane, Hadamard point (circle) joined to  $X$  point (square) for each rule that has a  $V$ ; the 81  $V$ -free rules are drawn as diamonds, one point each, because the gate cannot move them (Corollary 2). Almost every Hadamard point sits at  $(1, 2)$  — one giant sector — and the gate change fans them out to the right. Right: the sector-count base under one gate against the other. Nothing lies below the diagonal, which is Corollary 1 drawn.

## 4 The unitary / reversible baseline

The 16 rules built from  $I$  and  $V$  alone are the cleanest comparison available, for a specific reason: on the Hadamard side they are unitary, so sectors and monitored attractors coincide (R9's check A2), and on the  $X$  side they are reversible, so sectors and cycles coincide (R10's F1). The same object is being measured in both families, and the comparison cannot be confounded by the transient structure that dominates everywhere else.

| rule | tuple | Hadamard |        | $X$    |        | sector-count class |             |
|------|-------|----------|--------|--------|--------|--------------------|-------------|
|      |       | $a$      | $b$    | $a$    | $b$    | $H$                | $X$         |
| 51   | VVVV  | 1.0000   | 2.0000 | 2.0000 | 1.0000 | constant           | exponential |
| 54   | IVVV  | 1.0000   | 2.0000 | 2.0000 | 1.0000 | constant           | exponential |
| 57   | VIVV  | 1.0000   | 2.0000 | 1.3994 | 2.0000 | constant           | exponential |
| 60   | IIVV  | 1.0000   | 2.0000 | 2.0000 | 1.0000 | polynomial         | exponential |
| 99   | VVIV  | 1.0000   | 2.0000 | 1.3994 | 2.0000 | constant           | exponential |
| 102  | IVIV  | 1.0000   | 2.0000 | 2.0000 | 1.0000 | polynomial         | exponential |
| 105  | VIIIV | 1.0000   | 2.0000 | 2.0000 | 1.0000 | polynomial         | exponential |
| 108  | IIIV  | 1.7549   | 1.6180 | 1.8278 | 1.0000 | exponential        | exponential |
| 147  | VVVI  | 1.0000   | 2.0000 | 2.0000 | 1.0000 | constant           | exponential |
| 150  | IVVI  | 1.0000   | 2.0000 | 2.0000 | 1.0000 | polynomial         | exponential |
| 153  | VIVI  | 1.0000   | 2.0000 | 1.8311 | 1.0000 | constant           | exponential |
| 156  | IIVI  | 1.6180   | 1.3195 | 2.0000 | 1.2983 | exponential        | exponential |
| 195  | VVII  | 1.0000   | 2.0000 | 1.8311 | 1.0000 | constant           | exponential |
| 198  | IVII  | 1.6180   | 1.3195 | 2.0000 | 1.2983 | exponential        | exponential |
| 201  | VIII  | 1.7549   | 1.6180 | 1.8470 | 1.2339 | exponential        | exponential |
| 204  | IIII  | 2.0000   | 1.0000 | 2.0000 | 1.0000 | exponential        | exponential |

Three things stand out, and none of them is a small effect.

**The gate nearly reflects the family.** Eleven of the 16 sit at  $(a, b) = (1, 2)$  under the Hadamard — a single sector of size  $2^N$ , no fragmentation whatever. Under  $X$ , nine of those

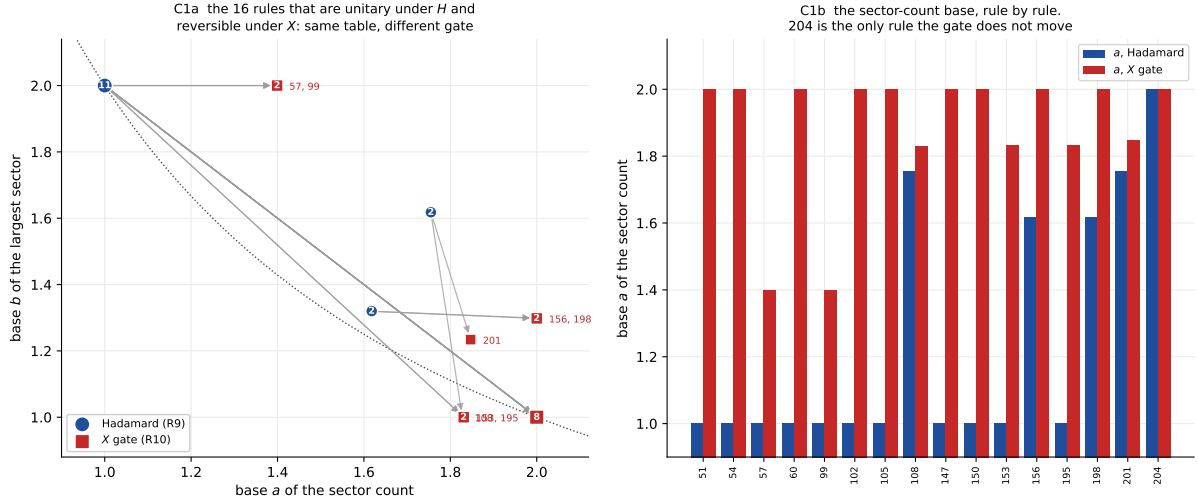


Figure 2: **C1, the baseline.** Left: each of the 16 rules under both gates, joined by the move; marker area counts rules sharing a cell. Right: the sector-count base rule by rule. Rule 204 is the only one the gate leaves where it was.

eleven end at  $b = 1$ : the largest sector is no longer a fixed fraction of the space at all. Seven go to  $(2, 1)$  exactly — exponentially many tiny orbits, maximal fragmentation — and 153 and 195 to  $(1.83, 1)$ . Only 57 and 99 keep  $b = 2$ , moving left instead of down. The extreme case is rule 51 (VVVV). Under the Hadamard it is the most ergodic rule in the family, one sector covering the whole basis. Under  $X$  it is  $x \mapsto \bar{x}$ , so the space splits into  $2^{N-1}$  two-cycles. Same symbol table, opposite ends of the sector axis, and Theorem 1 says which way round it must be: the Hadamard picture is the  $X$  picture after gluing.

**Only 204 is fixed.** Rule 204 is IIII, the identity, which contains no V and therefore cannot move (Corollary 2). It is the only one of the 16 with that property, and correspondingly the only rule whose  $a$  is identical in the two columns.

**The four fragmented rules stay fragmented, with different constants.** 156, 198, 108, 201 are the four rules whose sector count grows exponentially under the Hadamard, and all four still do under  $X$  — but the constants have no relation:

|          | Hadamard             | $X$                          |
|----------|----------------------|------------------------------|
| 156, 198 | $(\varphi, 4^{1/5})$ | $(2, 1.298)$                 |
| 108, 201 | $(1.7549, \varphi)$  | $(1.828, 2), (1.847, 1.234)$ |

The Hadamard values are exact algebraic numbers with derivations behind them (R2 §3's room-packing saddle point for  $4^{1/5}$ ; the golden ratio and the root of  $x^3 = 2x^2 - x + 1$  from integer recurrences). The  $X$ -gate values are not the same numbers and are not obviously algebraic. Fragmentation transfers; the *grammar* producing it does not.

**And the correlation is nil.**

| subset                    | coordinate           | rules | Pearson $r$ | mean $ \Delta $ | unchanged |
|---------------------------|----------------------|-------|-------------|-----------------|-----------|
| all 256                   | $a$ (sector count)   | 249   | +0.642      | 0.100           | 199       |
| all 256                   | $b$ (largest sector) | 251   | +0.561      | 0.095           | 202       |
| the 175 with a V          | $a$ (sector count)   | 170   | +0.465      | 0.146           | 120       |
| the 175 with a V          | $b$ (largest sector) | 172   | +0.334      | 0.138           | 127       |
| the 16 unitary/reversible | $a$ (sector count)   | 16    | +0.171      | 0.649           | 1         |
| the 16 unitary/reversible | $b$ (largest sector) | 16    | +0.034      | 0.628           | 3         |
| the 240 others            | $a$ (sector count)   | 233   | +0.759      | 0.062           | 198       |
| the 240 others            | $b$ (largest sector) | 235   | +0.698      | 0.058           | 199       |

On the 16 baseline rules the Pearson correlation between the two gates is +0.17 in  $a$  and +0.03 in  $b$ , with mean absolute differences of 0.65 and 0.63 on axes of total width 1. Read the “all 256” row with care: its apparent  $r = +0.64$  is inflated by the 81 rules that are the same circuit under both gates and therefore correlate perfectly by construction. Restricted to the 175 rules that actually contain a V,  $r$  falls to +0.47 and +0.33. The gate does not preserve exponents.

## 5 R9’s headline sets under the other gate

R9 singled out three sets of rules. Each is asked here what it does under  $X$ .

|                                               |       | $H$ : sector count |             | $X$ : sector count |             | $X$ cycle   |
|-----------------------------------------------|-------|--------------------|-------------|--------------------|-------------|-------------|
| rule                                          | tuple | class              | at $N = 16$ | class              | at $N = 16$ | class       |
| <i>open-system fragmented (R9 §6.3)</i>       |       |                    |             |                    |             |             |
| 156                                           | IIVI  | exponential        | 2 584       | exponential        | 7 744       | exponential |
| 198                                           | IVII  | exponential        | 2 584       | exponential        | 7 744       | exponential |
| 140                                           | IIDI  | exponential        | 2 584       | exponential        | 2 584       | constant    |
| 206                                           | IEII  | exponential        | 2 584       | exponential        | 2 584       | constant    |
| 196                                           | IDII  | exponential        | 2 584       | exponential        | 2 584       | constant    |
| 220                                           | IIEI  | exponential        | 2 584       | exponential        | 2 584       | constant    |
| 108                                           | IIIV  | exponential        | 10 252      | exponential        | 16 560      | polynomial  |
| 201                                           | VIII  | exponential        | 5 842       | exponential        | 12 076      | exponential |
| <i>strictly linear sector count (R9 §6.4)</i> |       |                    |             |                    |             |             |
| 44                                            | IIDV  | polynomial         | 17          | exponential        | 595         | constant    |
| 100                                           | IDIV  | polynomial         | 19          | exponential        | 595         | constant    |
| 110                                           | IEIV  | polynomial         | 17          | exponential        | 149         | polynomial  |
| 124                                           | IIEV  | polynomial         | 17          | exponential        | 149         | polynomial  |
| 188                                           | IIVE  | polynomial         | 17          | polynomial         | 17          | polynomial  |
| 230                                           | IVIE  | polynomial         | 17          | polynomial         | 17          | polynomial  |
| <i>pinned frontier (R9 §6.5)</i>              |       |                    |             |                    |             |             |
| 44                                            | IIDV  | polynomial         | 17          | exponential        | 595         | constant    |
| 60                                            | IIVV  | polynomial         | 17          | exponential        | 4 116       | constant    |
| 100                                           | IDIV  | polynomial         | 19          | exponential        | 595         | constant    |
| 102                                           | IVIV  | polynomial         | 17          | exponential        | 4 116       | constant    |
| 110                                           | IEIV  | polynomial         | 17          | exponential        | 149         | polynomial  |
| 124                                           | IIEV  | polynomial         | 17          | exponential        | 149         | polynomial  |
| 188                                           | IIVE  | polynomial         | 17          | polynomial         | 17          | polynomial  |
| 230                                           | IVIE  | polynomial         | 17          | polynomial         | 17          | polynomial  |

### 5.1 The open-system-fragmented rules

R9’s central physical claim is that eight V+reset rules keep an exponentially growing sector count, and that because a sector is an enclosure — span of a weak component is invariant under every Kraus operator — this is fragmentation of an *open* quantum system rather than of a unitary circuit with noise added afterwards.

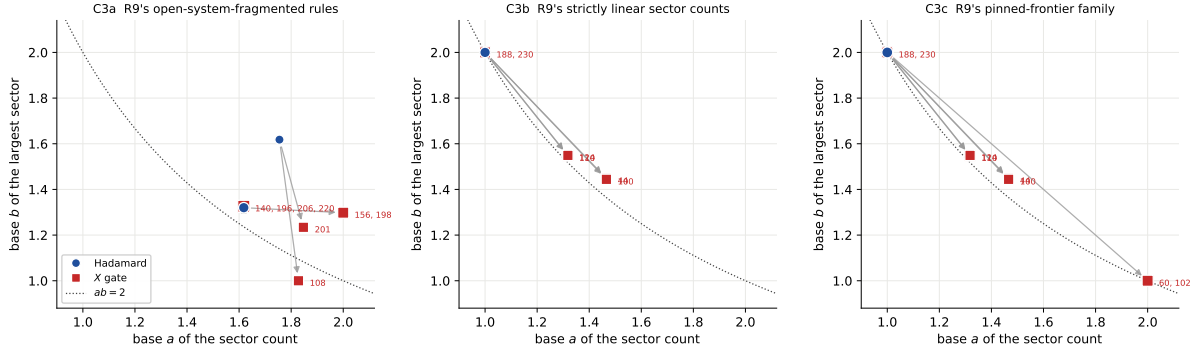


Figure 3: **C3, the three sets.** Hadamard (circle) to  $X$  (square) in the sector plane, common window. Labels list the rules sharing a destination cell.

All eight remain exponential under  $X$ , and four of them do something stronger: 140, 196, 206, 220 have *exactly* the same sector count as under the Hadamard, 2584 at  $N = 16$ , because they are  $V$ -free and hence the same circuit (Corollary 2). Their appearance in R9's list is therefore not a statement about the Hadamard at all: those four rules are classical, and their fragmentation is a classical fact that R9 measured with a quantum engine. The other four — 156, 198, 108, 201 — do contain a  $V$ , and for them the  $X$  count is strictly larger (7744 against 2584; 16560 against 10252; 12076 against 5842), exactly as Corollary 1 requires.

This is a sharpening of R9 rather than a contradiction of it. The claim “these rules fragment as open quantum systems” stands; what R11 adds is that for half of them the Hadamard was never doing any work.

## 5.2 The linear-sector-count rules and the pinned frontier

R9 §6.4 identified six rules — 44, 100, 110, 124, 188, 230 — whose sector count is strictly linear,  $N + 1$  of them, carried by a charge that is the position of the extremal excitation; §6.5 showed the charge is an obc0 artefact, collapsing to two sectors on the ring. Under  $X$  the six split three ways:

- 188 (IIVE) and 230 (IVIE) keep *exactly* 17 sectors at  $N = 16$ , the same linear count as under the Hadamard, though they contain a  $V$  and are therefore not forced to. For these two the pinned-frontier charge is gate-independent.
- 110 and 124 become exponential but only just, 149 sectors at  $N = 16$  against 17.
- 44 and 100 jump to 595.

The frontier family of §6.5 adds 60 and 102, and those two go all the way to 4116 sectors — from a linear count to  $a = 2$ , the maximum. So the pinned frontier is a Hadamard-specific structure for six of the eight rules that carry it, and survives the gate change for two. Combined with R9's finding that it does not survive the ring, the charge is doubly fragile: it depends on the boundary condition *and*, mostly, on the gate. That is a good reason not to build a physical story on it, and it is the sort of thing the comparison is for.

## 6 What each family says that the other cannot

The comparison is not symmetric, and the asymmetries are informative.

**Only the Hadamard family has interference.** Dark states, the distinction between the monitored and the unmonitored channel at the level of the support, and R7’s certificate all need a successor set with more than one element and amplitudes that can cancel. Under  $X$  there is a single Kraus branch on each basis state, so those objects are vacuous. The  $X$ -gate family is a *classical* control: it isolates what in R9 is graph combinatorics from what is quantum mechanics. The answer, from §3, is that the sector axis is almost entirely the former.

**Only the permutation family has a clean recurrent structure.** With out-degree one,  $n_{\text{rec}} = n_{\text{wcc}}$  identically and every basin is a sector, so questions about attractors have exact answers (R10 §§5–8). In the Hadamard family the same questions are genuinely harder — rule 22 has three attractors inside two sectors — which is why R9 could assert A4 in one direction only. The permutation family is where the attractor-level statements can be proved.

**The exclusion curve belongs to both.**  $ab \geq 2$  is a theorem about partitions and applies wherever sectors partition the basis, so it binds both families identically, and neither produced a violation. It does *not* apply to cycles, and R10 §5 makes that precise: the recurrent set is not the whole space, so the analogous constraint degenerates.

## 7 Open items

1. **pbc.** Held for both families. The refinement theorem is proved without reference to the boundary, so it should carry over verbatim, and it will be worth checking that the ring does not break the 81-rule identity.
2. **The gluing map.** Theorem 1 says each Hadamard sector is a union of  $X$  sectors, but nothing here counts them: the natural next observable is the number of  $X$  sectors per Hadamard sector, whose growth would measure exactly how much the non-flip branch glues.
3. **A third gate.**  $X$  and  $H$  are the two extremes — a permutation and a maximally superposing gate. A partial rotation would interpolate, and the refinement theorem suggests the sector lattice would interpolate monotonically with it.
4. **The four classical rules in R9’s open-fragmented list.** 140, 196, 206, 220 deserve a purely combinatorial derivation of their 2584 sectors; nothing quantum is involved and the count is Fibonacci.

## 8 Reproduce

```
cd code
python -m qca_fragmentation.permutation.compare --bc obc0
python -m qca_fragmentation.permutation.compare_tables --bc obc0
python -m qca_fragmentation.permutation.compare_figures --bc obc0 --rebuild
python -m pytest qca_fragmentation/tests/test_gate_compare.py -q
```

Code: `permutation/compare.py` (the refinement checks, the common-window descriptors, the correlations), `permutation/compare_tables.py`, `permutation/compare_figures.py`.  
 Data: `results_wcc/` and `results_xgate/`, `analytics/gate_compare_obc0.json`.